

Cervical Spine MRI Analysis Pipeline

Measurement Feature — Component Breakdown

Based on AUBMC Radiologist Consultation

Architecture Note: All measurements below are designed for a sagittal-only pipeline (one NIfTI in, one report out). Measurements requiring axial slices (canal transverse width, facet joint analysis) are deferred to future work, as supporting axial input would require the clinician to upload a second MRI file and the pipeline to match levels across two orientations.

Group 1: Vertebral Body Measurements

Requires: Vertebral body segmentation from TotalSpineSeg.

1.1 Vertebral Body AP Width

Anterior-posterior dimension of each vertebral body (C2–C7) on the mid-sagittal slice. Measured as the distance between anterior and posterior midpoints of the vertebral body contour. 6 points method.

1.2 Vertebral Body SI Height

Superior-inferior dimension of each vertebral body. Measured as the distance between superior and inferior midpoints. Used downstream for disc height index calculation.

1.3 Spondylolisthesis (Vertebral Slippage)

Quantifies in millimeters how far one vertebral body is displaced anteriorly or posteriorly relative to the vertebra below it. Uses the vertebral body corner coordinates already extracted for AP width and SI height measurements.

- Report per level (e.g., “C4 on C5: 3mm anterolisthesis”)
- Grade using Meyerding classification:
 - Grade I: 0–25% slippage
 - Grade II: 25–50% slippage
 - Grade III: 50–75% slippage
 - Grade IV: 75–100% slippage

1.4 Vertebral Signal Intensity Analysis (Experimental)

The radiologist noted that Hounsfield Unit analysis is standard for detecting osteoporosis on CT. HU is not directly available on MRI, but T1/T2 signal intensity within the vertebral body mask can serve as a proxy for bone quality. Vertebrae with abnormally low T1 signal may indicate fatty infiltration or osteoporotic changes.

Flag as experimental in the report. Requires access to raw MRI signal values, not just segmentation masks.

Group 2: Disc Measurements

Requires: Disc segmentation from TotalSpineSeg (labels starting at 63).

2.1 Disc SI Height

Superior-inferior dimension of each intervertebral disc. The primary disc measurement for assessing disc space narrowing and degeneration.

2.2 Disc AP Width

Anterior-posterior dimension of each disc. Useful for assessing disc bulging or protrusion when compared against the vertebral body AP width.

2.3 Disc Height Index (Derived)

Calculated as: disc SI height ÷ adjacent vertebral body SI height. Normalizes disc height relative to the patient's own vertebral dimensions. A decrease in disc height index across levels can indicate progressive degeneration.

2.4 Disc Degeneration Grading (Experimental)

Pfirrmann grading system (Grades I–V) based on T2 signal intensity and disc structure:

- **Bright hyperintense signal**, Grade I:
- **Inhomogeneous signal**, Grade II:
- **Intermediate gray signal**, Grade III:
- **Hypointense dark signal**, Grade IV:
- **Collapsed disc space**, Grade V:

This is a signal-intensity analysis, not a geometric measurement. Requires raw MRI T2 values within the disc mask. Flag as experimental.

Group 3: Canal & Cord Measurements (Stenosis)

Requires: Canal segmentation (CSF = TotalSpineSeg label 2) + spinal cord segmentation (label 1).

3.1 Canal AP Diameter

Anterior-posterior dimension of the spinal canal at each level. The primary measurement for central canal stenosis. Clinical thresholds:

- Normal: >13mm
- Relatively narrow: 10–13mm
- Critically narrow: <10mm

3.2 Spinal Cord AP Diameter

Anterior-posterior dimension of the spinal cord at each level. Required for SAC and myelopathy assessment. Cord compression is indicated when the cord AP diameter is reduced relative to adjacent levels.

3.3 SAC — Space Available for the Cord (Derived)

Calculated as: canal AP diameter – spinal cord AP diameter. Quantifies the buffer space around the cord. A reduced SAC indicates the cord is at risk of compression even if the canal diameter alone appears borderline.

3.4 Torg-Pavlov Ratio (Derived)

Calculated as: canal AP diameter ÷ vertebral body AP width. Normalizes canal size relative to the patient's own vertebral dimensions.

- Normal: ≥ 0.8
- Stenosis suspected: < 0.8

3.5 Most Stenotic Level Identification

Scans all measured levels and reports which has the minimum canal AP diameter. Surfaces the clinically most significant level without requiring the physician to scan all numbers manually.

Group 4: Alignment & Curvature

Requires: Vertebral body segmentation + new geometric approach for angle calculation.

4.1 C2–C7 Cobb Angle

Measures overall cervical sagittal alignment. Calculated as the angle between the inferior endplate of C2 and the inferior endplate of C7 (or superior endplate of C7, depending on convention — literature review needed).

4.2 Lordosis Classification

Interprets the Cobb angle into clinical categories:

- **~20–40° (literature varies)** Normal lordosis:
- **~0°** Straightened / loss of lordosis:
- **negative angle** Kyphotic (reversed curve):

Clinical significance: loss of lordosis can indicate muscle spasm (acute finding) or advanced degenerative disease (chronic finding). Significant kyphosis threshold $\sim 25^\circ$ (needs literature confirmation).

4.3 Segmental Angles

Angle between adjacent vertebrae (e.g., C3–C4, C4–C5). Identifies where curvature changes occur and can detect focal kyphosis at a single level even when overall alignment appears normal.

Group 5: Abnormal Finding Detection (Signal-Based)

Requires: Raw MRI signal analysis within segmentation masks. These are anomaly detection tasks, qualitatively different from the geometric measurements in Groups 1–4.

5.1 Myelomalacia Detection

Abnormal high T2 signal within the spinal cord, indicating cord damage or gliosis. Clinically significant because it suggests irreversible cord injury and worsens the prognosis for myelopathy patients. Detection approach: compare T2 signal intensity within the cord mask against a threshold or against the patient's own cord signal at unaffected levels.

5.2 Fracture Detection

Vertebral body shape deformity: compression fractures (loss of height), burst fractures (expansion), wedge deformity. Could be approached geometrically (comparing expected vs. actual vertebral body shape) or via signal changes (bone marrow edema on T2/STIR).

5.3 Tumor / Mass Detection

Abnormal signal regions within or adjacent to the spinal structures. Extremely difficult to do reliably with segmentation alone. For a course project, this is best scoped as “flagged regions with abnormal signal for physician review” rather than classification.

5.4 Post-Surgical Scar Detection

The radiologist mentioned that MRI can show scarring on the spine from prior surgery. Scar tissue has characteristic signal patterns (typically enhances with gadolinium contrast). Detection would depend on whether contrast-enhanced sequences are available in the input.

Recommendation for course scope: Group 5 features should be framed as “abnormal signal regions flagged for physician review” rather than definitive classifications. These are hard problems that could each be their own research project.

Group 6: Clinical Interpretation Layer

Not new measurements — combines outputs from Groups 1–5 into clinically meaningful flags and structured reports.

6.1 Radiculopathy Indicators

Flag findings consistent with nerve root compression: disc measurements suggesting herniation, reduced foraminal dimensions at specific levels. The report would note the pattern without diagnosing.

6.2 Myelopathy Indicators

Flag findings consistent with spinal cord compression: central canal stenosis + reduced SAC + cord signal changes (if Group 5 is implemented). Combine thresholds across multiple measurements.

6.3 Per-Level Structured Report

Compile all measurements for each level (C2–C3 through C6–C7) into a radiology-style findings section. Each level gets: vertebral body dimensions, disc dimensions, canal diameter, cord diameter, SAC, Torg-Pavlov ratio, and any flags.

6.4 Demographic Percentile Comparison

Use quantile regression models (trained on Duke CSpineSeg dataset with age and sex as inputs) to report where the patient's measurements fall relative to the population. Example output: "Canal AP at C5: 11.2mm (12th percentile for 55-year-old male)."

Deferred: Requires Axial Input

The following measurements were mentioned by the radiologist but require axial MRI slices, which are a separate file from the sagittal series. Supporting these would require the clinician to upload two files and the pipeline to match vertebral levels across two orientations. Deferred to future work.

Canal Transverse (ML) Width

Mediolateral dimension of the canal. Requires axial slice.

Facet Joint Analysis

- Facet joint space measurement (gap between articular surfaces)
- Facet asymmetry (compare left vs. right at same level to detect arthritis/degeneration)

Facet joints are also not a standard TotalSpineSeg output, so this would require new segmentation work in addition to axial input.

Suggested Build Order

Phase	Components
Phase 1	Foundation: Mid-sagittal slice selection 1.1 Vertebral body AP width 1.2 Vertebral body SI height
Phase 2	3.1 Canal AP diameter 3.4 Torg-Pavlov ratio (derived from 1.1 + 3.1) 3.5 Most stenotic level identification
Phase 3	2.1 Disc SI height 2.2 Disc AP width 2.3 Disc height index (derived)
Phase 4	3.2 Spinal cord AP diameter 3.3 SAC (derived from 3.1 + 3.2)
Phase 5	4.1 C2–C7 Cobb angle 4.2 Lordosis classification 4.3 Segmental angles 1.3 Spondylolisthesis
Phase 6	6.1–6.2 Clinical syndrome flagging 6.3 Per-level structured report 6.4 Demographic percentile comparison
Phase 7	5.1 Myelomalacia detection (experimental) 2.4 Disc degeneration grading (experimental) 1.4 Vertebral signal intensity (experimental) 5.2–5.4 Fracture / tumor / scar detection (experimental)

Open Questions

Literature Review Needed

- Exact C2–C7 Cobb angle thresholds for lordosis classification (normal range, significant kyphosis cutoff)
- Cobb angle convention: inferior endplate of C2 to inferior vs. superior endplate of C7
- Pfirrmann grading thresholds for automated classification
- Meyerding grading: how to calculate % slippage from corner coordinates

Scope Decisions

- Group 5 (signal-based findings): include as experimental, or defer entirely?
- Group 6 (interpretation layer): how much clinical reasoning should the pipeline attempt vs. just reporting numbers?
- Vertebral signal intensity for osteoporosis: viable from MRI alone, or too unreliable without CT?